

JEE Mains 2019 Chapter wise Question Bank

Motion in Two Dimension - Questions

Q1

A particle is moving with a velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is a constant. The general equation for its path is:

- (1) $y = x^2 + \text{constant}$ (2) $y^2 = x + \text{constant}$
 (3) $y^2 = x^2 + \text{constant}$ (4) $xy = \text{constant}$

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Q2

The position co-ordinates of a particle moving in a 3-D coordinate system is given by

$$x = a \cos \omega t$$

$$y = a \sin \omega t$$

and $z = a\omega t$

The speed of the particle is:

- (1) $\sqrt{2} a\omega$ (2) $a\omega$
 (3) $\sqrt{3} a\omega$ (4) $2a\omega$

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Q3

Two guns A and B can fire bullets at speeds 1 km/s and 2 km/s respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is:

- (1) 1 : 16 (2) 1 : 2 (3) 1 : 4 (4) 1 : 8

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Q4

Two forces P and Q, of magnitude 2F and 3F, respectively, are at an angle θ with each other. If the force Q is doubled, then their resultant also gets doubled. Then, the angle θ is:

- (1) 120° (2) 60°
 (3) 90° (4) 30°

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Q5

A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Considering only translational and rotational modes, the total internal energy of the system is :

- (1) 15 RT (2) 12 RT
 (3) 4 RT (4) 20 RT

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Q6

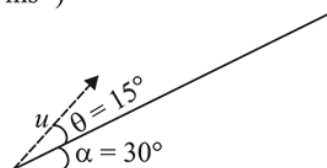
The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight?

- (1) 90° (2) 150° (3) 120° (4) 60°

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Q7

A plane is inclined at an angle $\alpha = 30^\circ$ with respect to the horizontal. A particle is projected with a speed $u = 2 \text{ ms}^{-1}$, from the base of the plane, as shown in figure. The distance from the base, at which the particle hits the plane is close to : (Take $g = 10 \text{ ms}^{-2}$)



- (1) 20 cm (2) 18 cm (3) 26 cm (4) 14 cm

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Q8

A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product $t_1 t_2$ is :

- (1) $R/4g$ (2) R/g (3) $R/2g$ (4) $2R/g$

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Q9

The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle θ_0 with speed v_0 then ($g = 10 \text{ ms}^{-2}$):

(1) $\theta_0 = \sin^{-1} \frac{1}{\sqrt{5}}$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$

(2) $\theta_0 = \cos^{-1} \left(\frac{2}{\sqrt{5}} \right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$

(3) $\theta_0 = \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$

(4) $\theta_0 = \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$

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Q10

Two particles are projected from the same point with the same speed u such that they have the same range R , but different maximum heights, h_1 and h_2 . Which of the following is correct ?

(1) $R^2 = 4 h_1 h_2$ (2) $R^2 = 16 h_1 h_2$

(3) $R^2 = 2 h_1 h_2$ (4) $R^2 = h_1 h_2$

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Motion in Two Dimension - Answers

Q1

(3) From given equation,

$$\vec{V} = K(y\hat{i} + x\hat{j})$$

$$\frac{dx}{dt} = ky \text{ and } \frac{dy}{dt} = kx$$

$$\text{Now, } \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{x}{y} = \frac{dy}{dx} \Rightarrow ydy = xdx$$

Integrating both side

$$y^2 = x^2 + c$$

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Q2

(1) Here, $v_x = -a\omega \sin \omega t$, $v_y = a\omega \cos \omega t$
and $v_z = a\omega$

$$\Rightarrow v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\Rightarrow v = \sqrt{(-a\omega \sin \omega t)^2 + (a\omega \cos \omega t)^2 + (a\omega)^2}$$

$$v = \sqrt{2}a\omega$$

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Q3

(1) As we know, range $R = \frac{u \sin 2\theta}{g}$ and, area $A = \pi R^2$

$$\therefore A \propto R^2 \text{ or, } A \propto u^4$$

$$\therefore \frac{A_1}{A_2} = \frac{u_1^4}{u_2^4} = \left[\frac{1}{2}\right]^4 = \frac{1}{16}$$

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Q4

(1) Using, $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

$$4F^2 + 9F^2 + 12F^2 \cos \theta = R^2$$

When forces Q is doubled,

$$4F^2 + 36F^2 + 24F^2 \cos \theta = 4R^2$$

$$4F^2 + 36F^2 + 24F^2 \cos \theta$$

$$= 4(13F^2 + 12F^2 \cos \theta) = 52F^2 + 48F^2 \cos \theta$$

$$\therefore \cos \theta = -\frac{12F^2}{24F^2} = -\frac{1}{2} \Rightarrow \theta = 120^\circ$$

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Q5

$$(1) U = \frac{f_1}{2} n_1 RT + \frac{f_2}{2} n_2 RT$$

Considering translational and rotational modes,
degrees of freedom $f_1 = 5$ and $f_2 = 3$

$$\therefore u = \frac{5}{2}(3RT) + \frac{3}{2} \times 5RT$$

$$U = 15RT$$

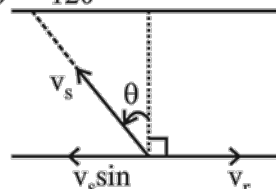
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Q6

(3) To cross the river straight

$$V_s \sin \theta = V_r \therefore \sin \theta = \frac{v_r}{v_s} = \frac{2}{4} = \frac{1}{2}$$

$$\therefore \theta = 30^\circ$$

Direction of swimmer with respect to flow
 $= 90^\circ + 30^\circ = 120^\circ$ 

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Q7

Motion in Two Dimension

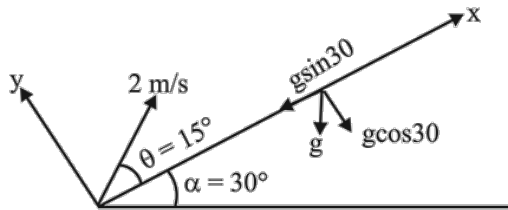
- (1) On an inclined plane, time of flight (T) is given by

$$T = \frac{2u \sin \theta}{g \cos \alpha}$$

Substituting the values, we get

$$T = \frac{(2)(2 \sin 15^\circ)}{g \cos 30^\circ} = \frac{4 \sin 15^\circ}{10 \cos 30^\circ}$$

$$\text{Distance, } S = (2 \cos 15^\circ)T - \frac{1}{2}g \sin 30^\circ(T)^2$$



$$= (2 \cos 15^\circ) \frac{4 \sin 15^\circ}{10 \cos 30^\circ} - \left(\frac{1}{2} \times 10 \sin 30^\circ \right) \frac{16 \sin^2 15^\circ}{100 \cos^2 30^\circ}$$

$$= \frac{16\sqrt{3} - 16}{60} \approx 0.1952 \text{m} \approx 20 \text{cm}$$

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Q8

- (4) R will be same for θ and $90^\circ - \theta$.

Time of flights:

$$t_1 = \frac{2u \sin \theta}{g} \text{ and}$$

$$t_2 = \frac{2u \sin(90^\circ - \theta)}{g} = \frac{2u \cos \theta}{g}$$

$$\text{Now, } t_1 t_2 = \left(\frac{2u \sin \theta}{g} \right) \left(\frac{2u \cos \theta}{g} \right)$$

$$= \frac{2}{g} \left(\frac{u^2 \sin 2\theta}{g} \right) = \frac{2R}{g}$$

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Q9

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- (3) Given, $y = 2x - 9x^2$

On comparing with,

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta},$$

We have,

$$\tan \theta = 2 \text{ or } \cos \theta = \frac{1}{\sqrt{5}}$$

$$\text{and } \frac{g}{2u^2 \cos^2 \theta} = 9 \text{ or } \frac{10}{2u^2 (1/\sqrt{5})^2} = 9$$

$$\therefore u = 5/3 \text{ m/s}$$

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Q10

- (2) For same range, the angle of projections are : θ and $90^\circ - \theta$. So,

$$h_1 = \frac{u^2 \sin^2 \theta}{2g} \text{ and}$$

$$h_2 = \frac{u^2 \sin^2(90^\circ - \theta)}{2g} = \frac{u^2 \cos^2 \theta}{2g}$$

$$\text{Also, } R = \frac{u^2 \sin 2\theta}{g}$$

$$h_1 h_2 = \frac{u^2 \sin^2 \theta}{2g} \times \frac{u^2 \cos^2 \theta}{2g}$$

$$= \frac{u^2}{16} \frac{u^2 (2 \sin \theta \cos \theta)^2}{g^2}$$

$$= \frac{R^2}{16}$$

$$\text{or } R^2 = 16 h_1 h_2$$

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